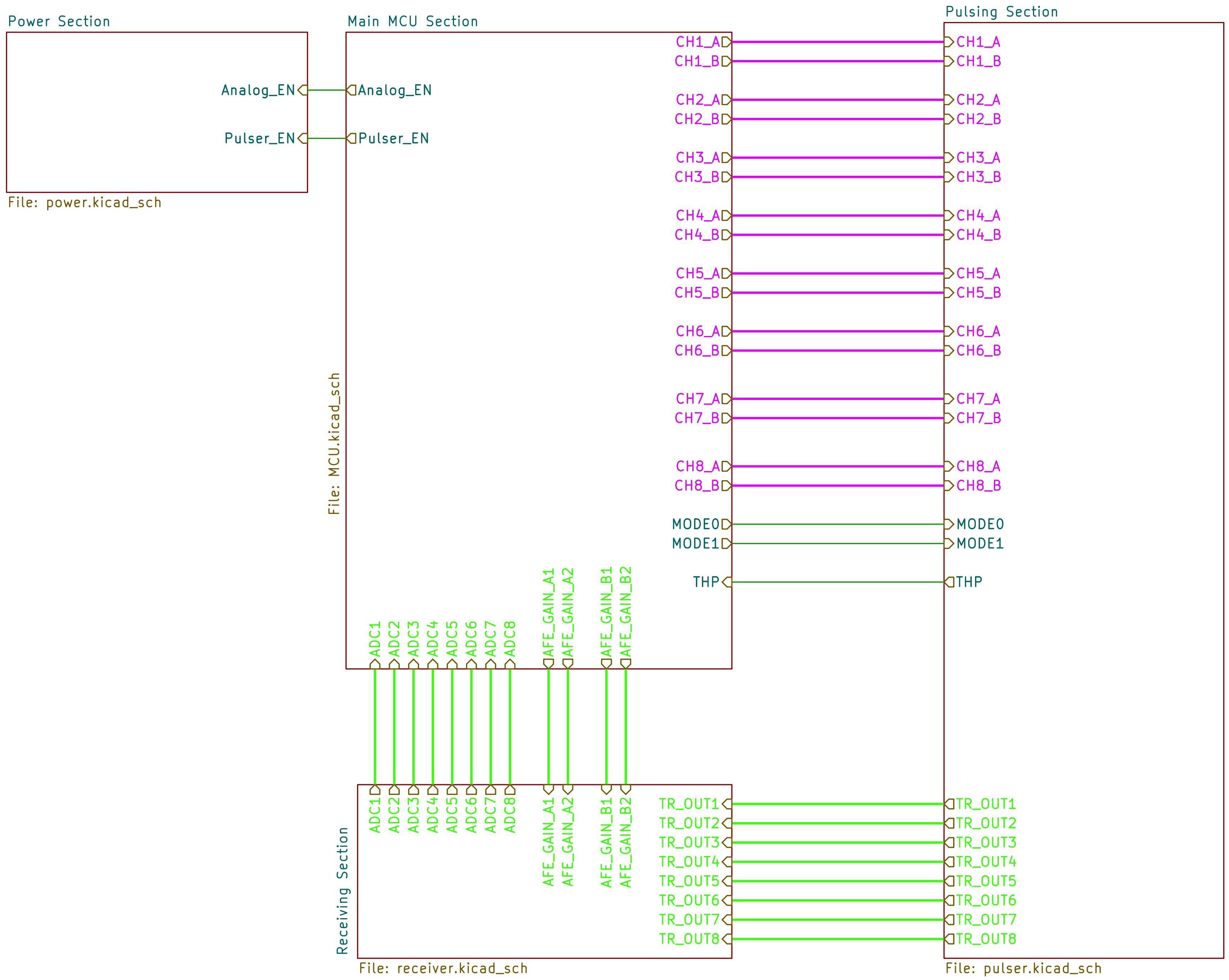


12 Volt (External) **(555mA)**	**7 Volt** (Buck Converter) **(350mA)**	**5 Volt** (LDO Regulator) **(200mA)**	LNA + VGA Supply **(160mA** – short circuit)
			Op-Amps (4 * 10mA = **40mA**)
		3.3 Volt (LDO Regulator) **(150mA)**	MCU Supply **(150mA)**
	5 Volt (Buck Converter) **(205mA)**	Positive Pulse Driver Supply **(100mA)**	
		3.3 Volt (LDO Regulator) **(5mA)**	Pulser Logic Supply **(5mA)**
		-5 Volt (Inverting Buck Converter) **(100mA)**	Negative Pulse Driver Supply **(100mA)**

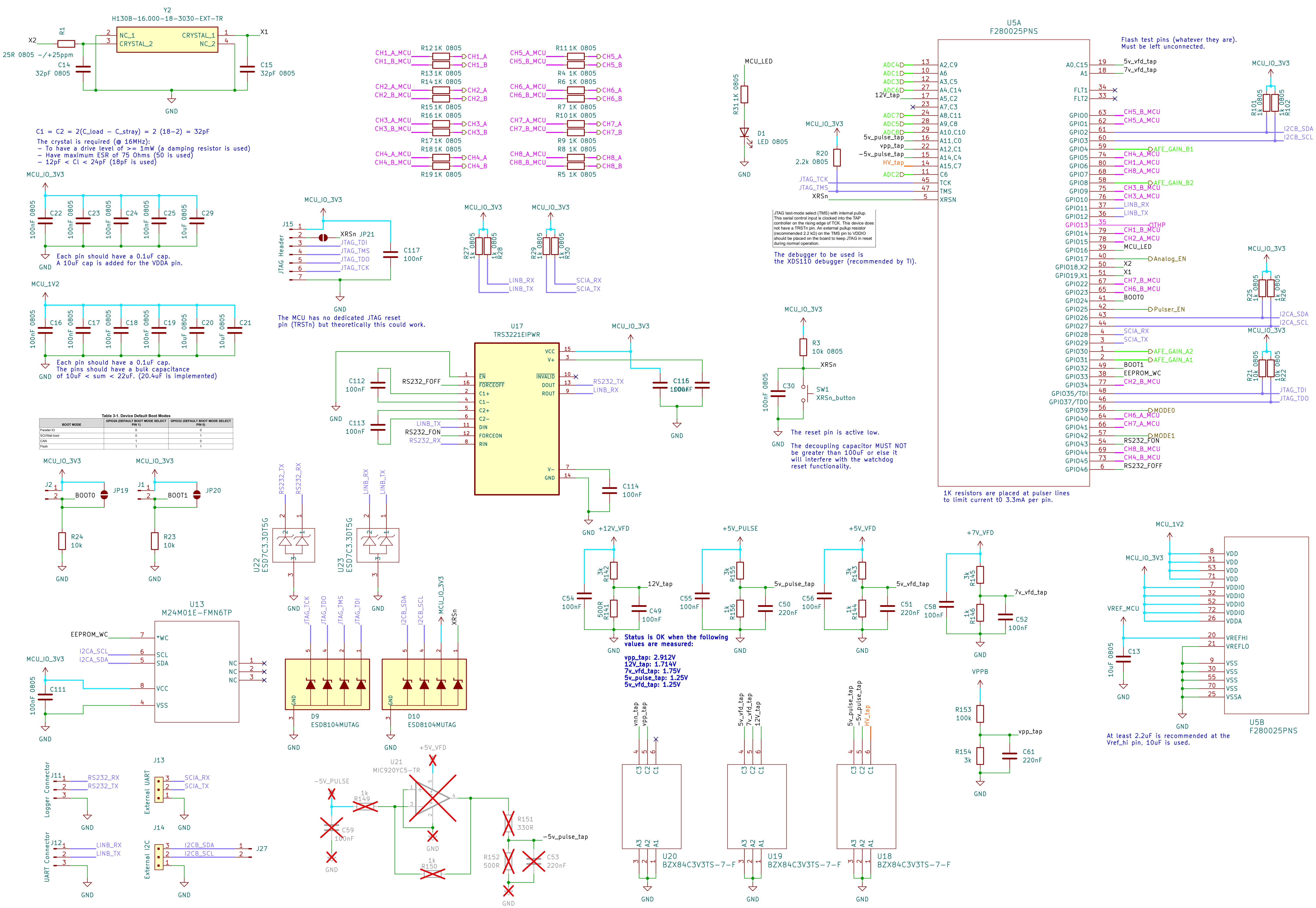
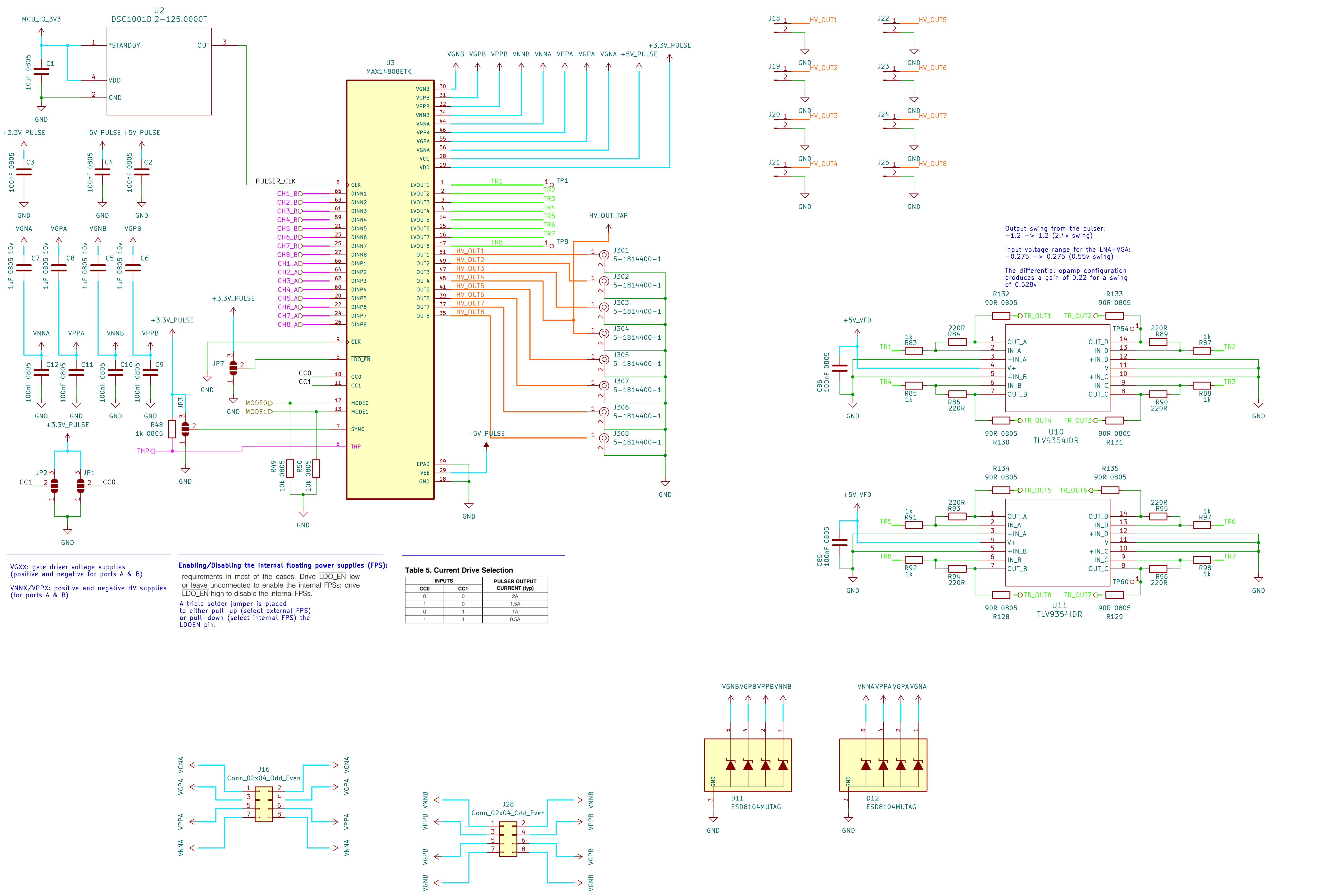


Table 3-1. Device Default Boot Modes

BOOT MODE	GPIO24 (DEFAULT BOOT MODE SELECT PIN 1)	GPIO32 (DEFAULT BOOT MODE SELECT PIN 0)
Parallel IO	0	0
SCI/Wait boot	0	1
CAN	1	0
Flash	1	1



VGXX: gate driver voltage supplies (positive and negative for ports A & B)

VNNX/VPPX: positive and negative HV supplies (for ports A & B)

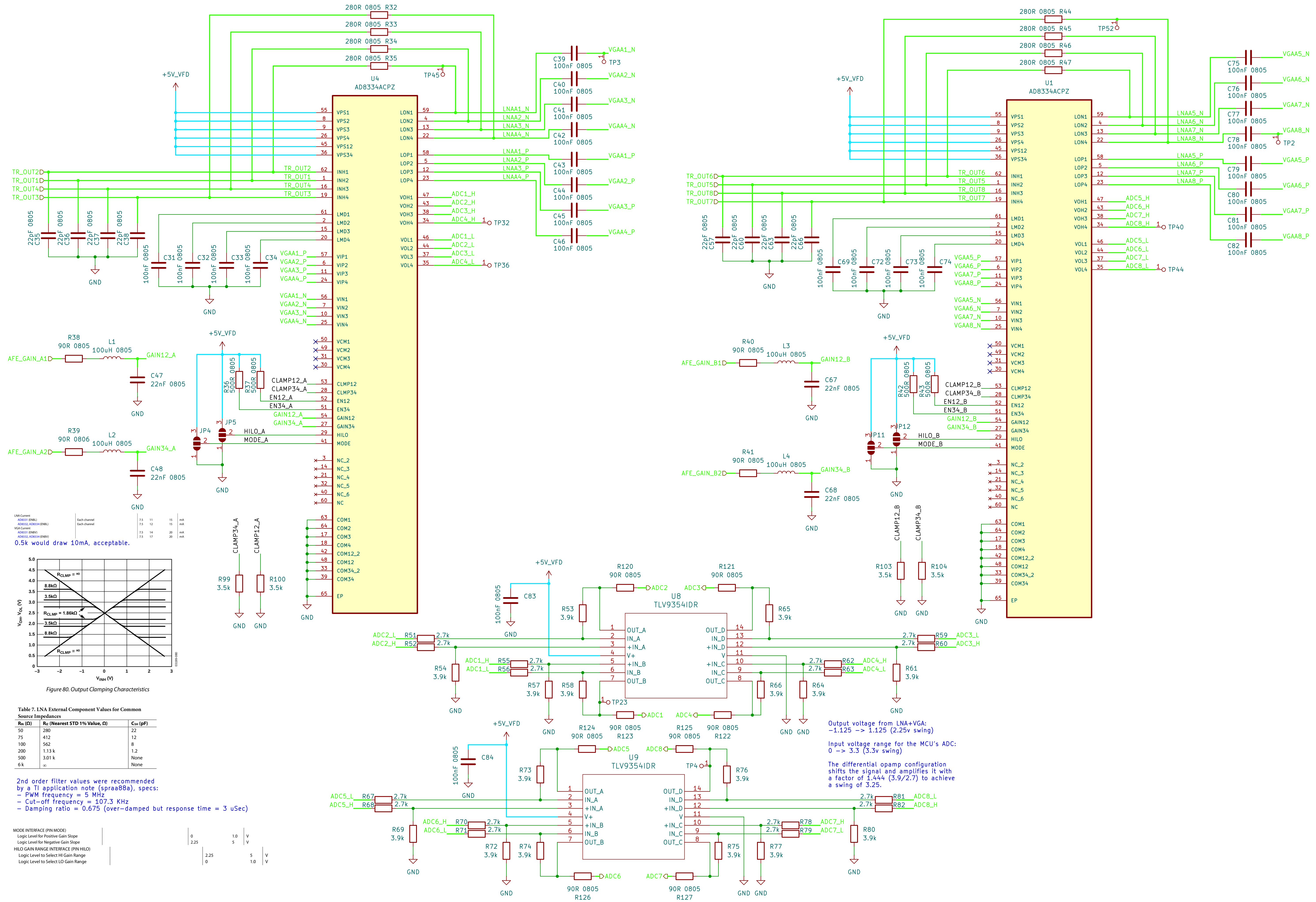
Enabling/Disabling the internal floating power supplies (FPS):

requirements in most of the cases. Drive LDO_EN low or leave unconnected to enable the internal FPSs; drive LDO_EN high to disable the internal FPSs.

A triple solder jumper is placed to either pull-up (select external FPS) or pull-down (select internal FPS) the LDOEN pin.

Table 5. Current Drive Selection

INPUTS		PULSER OUTPUT CURRENT (typ)
CC0	CC1	
0	0	2A
1	0	1.5A
0	1	1A
1	1	0.5A



LNA Current

LNA Current					
AD8331 (ENBL)	Each channel	7.5	11	15	mA
AD8332, AD8334 (ENBL)	Each channel	7.5	12	15	mA
VGA Current					
AD8331 (ENBV)		7.5	14	20	mA
AD8332, AD8334 (ENBV)		7.5	17	20	mA

0.5k would draw 10mA, acceptable.

Figure 80. Output Clamping Characteristics

Table 7. LNA External Component Values for Common Source Impedances

R_{IN} (Ω)	R_Z (Nearest STD 1% Value, Ω)	C_{SH} (pF)
50	280	22
75	412	12
100	562	8
200	1.13 k	1.2
500	3.01 k	None
6 k	∞	None

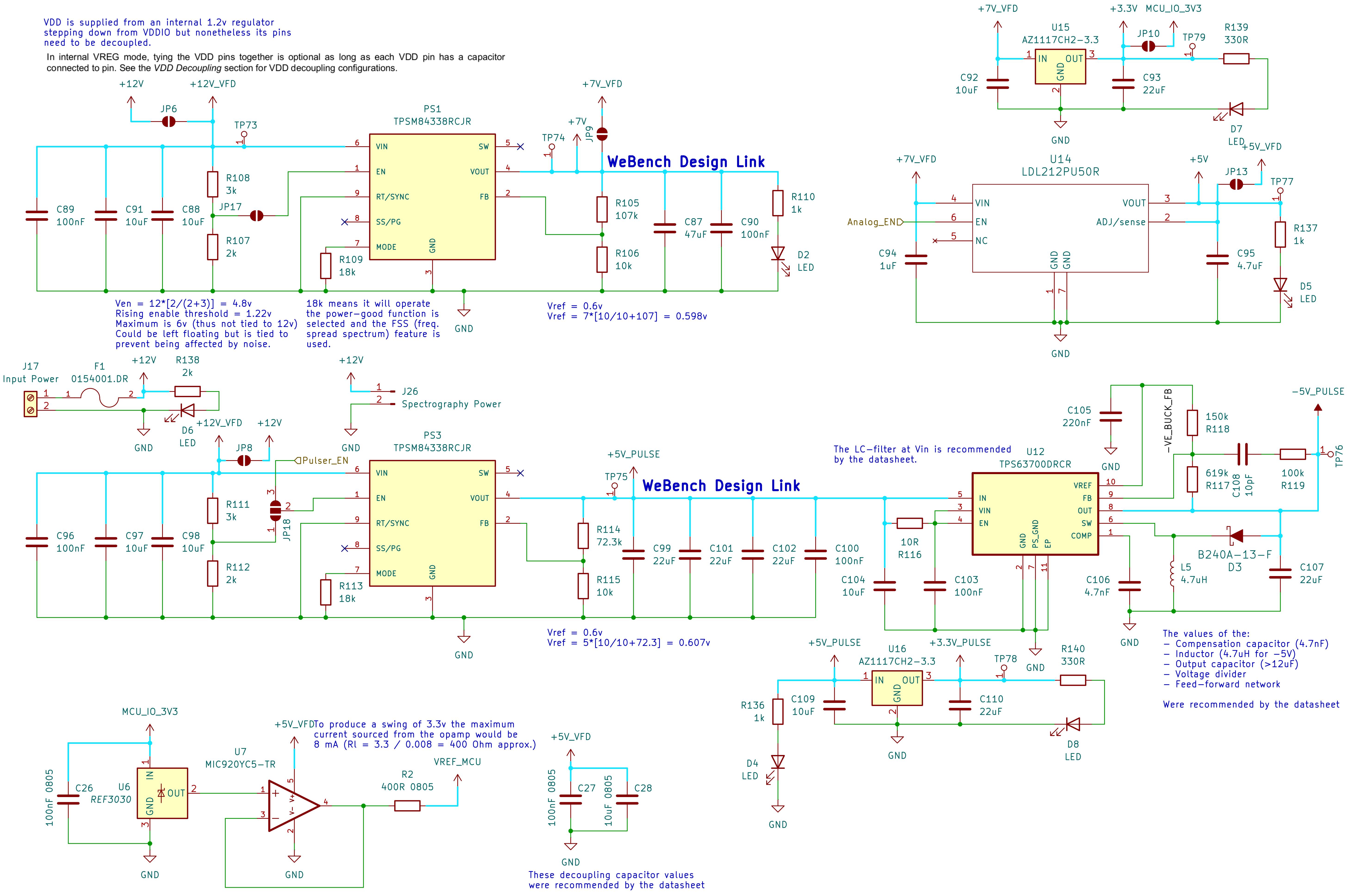
2nd order filter values were recommended
by a TI application note (spraa88a), specs:
- PWM frequency = 5 MHz
- Cut-off frequency = 107.3 KHz
- Damping ratio = 0.675 (over-damped but response time = 3 uSec)

MODE INTERFACE (PIN MODE)				
Logic Level for Positive Gain Slope		0	1.0	V
Logic Level for Negative Gain Slope		2.25	5	V
HILO GAIN RANGE INTERFACE (PIN HILO)				
Logic Level to Select HI Gain Range		2.25	5	V
Logic Level to Select LO Gain Range		0	1.0	V

Output voltage from LNA+VGA:
-1.125 -> 1.125 (2.25v swing)

Input voltage range for the MCU's ADC:
0 -> 3.3 (3.3v swing)

The differential opamp configuration
shifts the signal and amplifies it with
a factor of 1.444 (3.9/2.7) to achieve
a swing of 3.25.


VDD is supplied from an internal 1.2v regulator stepping down from VDDIO but nonetheless its pins need to be decoupled.
In internal VREG mode, tying the VDD pins together is optional as long as each VDD pin has a capacitor connected to pin. See the *VDD Decoupling* section for VDD decoupling configurations.
+12V
+12V_VFD
JP6
TP73
PS1
TPSM84338RCJR
VIN
EN
RT/SYNC
SS/PG
MODE
GND
SW
VOUT
FB
+7V_VFD
+7V
TP74
JP9
WeBench Design Link
C89 100nF
C91 10uF
C88 10uF
R108 3k
JP17
R107 2k
R109 18k
R105 107k
R106 10k
C87 47uF
C90 100nF
R110 1k
D2 LED
GND
Ven = 12*[2/(2+3)] = 4.8v
Rising enable threshold = 1.22v
Maximum is 6v (thus not tied to 12v)
Could be left floating but is tied to prevent being affected by noise.
18k means it will operate the power-good function is selected and the FSS (freq. spread spectrum) feature is used.
Vref = 0.6v
Vref = 7*[10/10+107] = 0.598v
+7V_VFD
U15
AZ1117CH2-3.3
IN
OUT
GND
+3.3V
MCU_IO_3V3
JP10
TP79
R139 330R
C92 10uF
C93 22uF
D7 LED
U14
LDL212PU50R
VIN
EN
NC
VOUT
ADJ/sense
+5V
JP13
+5V_VFD
TP77
Analog_EN
C94 1uF
C95 4.7uF
R137 1k
D5 LED
J17 Input Power
F1 0154001.DR
R138 2k
D6 LED
+12V
J26 Spectrography Power
+12V_VFD
JP8
Pulser_EN
PS3
TPSM84338RCJR
JP18
+5V_PULSE
TP75
WeBench Design Link
C96 100nF
C97 10uF
C98 10uF
R111 3k
R112 2k
R113 18k
R114 72.3k
R115 10k
C99 22uF
C101 22uF
C102 22uF
C100 100nF
Vref = 0.6v
Vref = 5*[10/10+72.3] = 0.607v
The LC-filter at Vin is recommended by the datasheet.
U12
TPS63700DRCR
IN
VIN
EN
VREF
FB
OUT
SW
COMP
PS_GND
EP
C105 220nF
-VE_BUCK_FB
R118 150k
R117 619k
C108 10pF
R119 100k
-5V_PULSE
TP76
10R R116
C104 10uF
C103 100nF
C106 4.7nF
L5 4.7uH
B240A-13-F D3
C107 22uF
The values of the:
- Compensation capacitor (4.7nF)
- Inductor (4.7uH for -5V)
- Output capacitor (>12uF)
- Voltage divider
- Feed-forward network
Were recommended by the datasheet
+5V_PULSE
U16
AZ1117CH2-3.3
+3.3V_PULSE
TP78
R140 330R
R136 1k
C109 10uF
C110 22uF
D8 LED
D4 LED
MCU_IO_3V3
C26 100nF 0805
U6 REF3030
U7 MIC920YC5-TR
+5V_VFD
To produce a swing of 3.3v the maximum current sourced from the opamp would be 8 mA (Rl = 3.3 / 0.008 = 400 Ohm approx.)
R2 400R 0805
VREF_MCU
+5V_VFD
C27 100nF 0805
C28 10uF 0805
These decoupling capacitor values were recommended by the datasheet